Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

$NaOH$ cannot be used in place of NH_4Cl and NH_4OH for the precipitation of III group radicals, since:

(A) Nickel will be precipitated

(B) Concentration of hydroxide ions is very low

(C) Sodium ions will react with acidic radicals

(D) Na^+ ions interfere with the detection of III group

Q.2

Identify the correct order of solubility of $Na_2\ S, CuS$ and ZnS in aqueous medium

(A) $CuS > ZnS > Na_2\ S$

(B) $ZnS > Na_2\ S > CuS$

(C) $Na_2\ S > CuS > ZnS$

(D) $Na_2\ S > ZnS > CuS$

Q.3

A solution of a metal ion, when treated with KI, gives a red precipitate which dissolves in an excess of KI to give a colourless solution. However the solution of the metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is

(A) Pb^{2+}

(B) Cu^{2+}

(C) Hg^{+2}

(D) CO^{2+}

Q.4

Which of the following gives black precipitate when $H_2\ S$ gas is passed through its solution

(A) Acidic $AgNO_3$

(B) $Mg\ (NO_3)_2$

(C) $Ammonical BaCl_2$

(D) Copper nitrate

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Q.5

A substance on treatment with dil H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of

(A) CO_3^{2-}

(B) S^{2-}

(C) SO_3^{2-}

(D) NO_2^-

Q.6

Which of the following is not a preliminary test used to detect ions?

(A) Borax bead test

(B) Flame test

(C) Brown ring test

(D) Microcosmic salt bead test

Q.7

Which of the following can be used to precipitate lead salt

(A) K_2CrO_4

(B) 2 KI

(C) dil. H_2SO_4

(D) All

Q.8

$ZnX \xrightarrow{\text{dil } H_2SO_4} \underset{\text{Colorless}}{A} + B$

$A \xrightarrow{K_2C_2O_7} Y$ (Yellow turbidity) + Green Solution.

Then X is

(A) NO_2^-

(B) $S_2O_3^{2-}$

(C) SO_3^{2-}

(D) S^{2-}

Q.9

Of the following sulphides which one is insoluble in dilute acids but soluble in alkalies

(A) PbS

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(B) CdS

(C) FeS

(D) Sb_2S_3

Q.10

$$PbCl_2 + KI \rightarrow \underset{\text{Yellow ppt}}{[A]} + KCl$$

$$\underset{\text{Yellow ppt}}{[A]} + \underset{\text{excess}}{KI} \longrightarrow \underset{\text{soluble}}{[B]}$$

Compound $[A]$ and $[B]$ are

(A) PbI_4 and $K_2[PbI_4]$ respectively

(B) $K_2[PbI_4]$ and PbI_4 respectively

(C) PbI_2 and $K_2[PbI_4]$ respectively

(D) PbI_2 and $K_2[PbI_2]$ respectively

Q.11

Fe^{2+} ion can be distinguished by Fe^{3+} ion by

(A) NH_4SCN

(B) $AgNO_3$

(C) $BaCl_2$

(D) None of these

Q.12

$AgCl$ dissolves in ammonia solution giving

(A) Ag^+, NH_4^+ and Cl^-

(B) $Ag(NH_3)^+$ and Cl^-

(C) $Ag_2(NH_3)^+$ and Cl^-

(D) $Ag(NH_3)_2^+$ and Cl^-

Q.13

[A solid compound X on heating gives CO_2 gas and a residue. The residue mixed with water forms $\mathbf{Y}$. On passing excess of CO_2, through $\mathbf{Y}$ in water, a clear solution Z is obtained. On boiling Z, compound X is reformed. The compound X is

(A) $Ca(HCO_3)_2$

(B) $CaCO_3$

(C) Na_2CO_3

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(D) K_2CO_3

Q.14

When $H_2\ S$ gas is passed into a certain solution, it reacts to form a white precipitate. The solution referred to contains ions of

(A) Lead

(B) Zinc

(C) Copper

(D) Nickel

Q.15

A precipitate of ..... would be obtained on adding HCl to a solution of $(Sb_2\ S_3)$ in yellow ammonium sulphide

(A) $Sb_2\ S_3$

(B) $Sb_2\ S_5$

(C) SbS

(D) SbS_2

Q.16

Mark the correct statement out of the following

(A) Group I basic radicals precipitate as chlorides

(B) Group IV basic radicals precipitate as sulphides

(C) Group V basic radicals precipitate as carbonates

(D) All of the above statements are correct

Q.17

Passage:
A metal sulphide (A) gives (B) gas with rotten egg smell and (C) a colourless sulphates, when treated with dil. H_2SO_4. (B) reacts with $K_2Cr_2O_7/H^+$ to form (D) a white grey element. (D) burns in oxygen to yield (E) a colourless gas. If (B) is added in (E), it gives (D) and colourless liquid, which turns anhydrous $CuSO_4$ blue. (C) gives a precipitate with $NH_{3\ (aq.)}$ or $NaOH$ which dissolves in excess of $NaOH$.
Question:
A is:

(A) CuS

(B) ZnS

(C) $Cu_2\ S$

(D) None of these

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Q.18

Passage:

A metal sulphide (A) gives (B) gas with rotten egg smell and (C) a colourless sulphates, when treated with dil. H_2SO_4. (B) reacts with $K_2Cr_2O_7/H^+$ to form (D) a white grey element. (D) burns in oxygen to yield (E) a colourless gas. If (B) is added in (E), it gives (D) and colourless liquid, which turns anhydrous $CuSO_4$ blue. (C) gives a precipitate with $NH_{3\ (aq.)}$ or $NaOH$ which dissolves in excess of $NaOH$.

Question:

B is:

(A) SO_2

(B) SO_3

(C) $H_2\ S$

(D) CS_2

Q.19

Passage:

A metal sulphide (A) gives (B) gas with rotten egg smell and (C) a colourless sulphates, when treated with dil. H_2SO_4. (B) reacts with $K_2Cr_2O_7/H^+$ to form (D) a white grey element. (D) burns in oxygen to yield (E) a colourless gas. If (B) is added in (E), it gives (D) and colourless liquid, which turns anhydrous $CuSO_4$ blue. (C) gives a precipitate with $NH_{3\ (aq.)}$ or $NaOH$ which dissolves in excess of $NaOH$.

Question:

C is:

(A) $CuSO_4$

(B) $BaSO_4$

(C) $ZnSO_4$

(D) Na_2ZnO_2

Q.20

Passage:

There are four bottles which contains :

Copper (II) sulphate Lead nitrate

Hydrochloric acid Sodium carbonate

By mixing samples of the contents of the bottles in pairs, a chemist made the following observations:

Bottle (1) + Bottle (2) white precipitate

Bottle (1) + Bottle (4) white precipitate

Bottle (2) + Bottle (4) no visible reaction

Bottle (1) + Bottle (3) white precipitate

Bottle (2) + Bottle (3) colourles gas evolved

Bottle (3) + Bottle (4) green blue precipitate

Question:

Bottle (1) contains :

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(A) $Pb(NO_3)_2$

(B) HCl

(C) Na_2CO_3

(D) $CuSO_4$

Answers & Solutions

Q.1 Answer:

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Solution:

Na_2CO_3 **will yield large concentration of ions to exceed the solubility product of** $MgCO_3$ **and it will be precipitated along with carbonates of** Ca^2, Sr^2 **and** Ba^{2*} **in fifth group. Further, in absence of common ion** NH_4OH **will ionize more to give sufficient** $\left[OH^-\right]$ **to exceed the solubility product of** $Mg(OH)$, **and its ppt.**

Q.2 Answer:

$Na_2\ S > ZnS > CuS$

Solution:

Ionic compounds are more soluble in water or in aqueous medium.

According to Fajans' rule,

Size of the cation increases the ionic character also increases.

Ionic character oc size of cation (if anion is

same)

The order of size of cation is

$Na^+ > Zn^{2+} > Cu^{2+}$

∴ The order of ionic character and hence, of solubility in water is as

$Na_2S > ZnS > CuS$

Q.3 Answer:

Hg^{+2}

Solution:

$$Hg^{+2} + 2I^- \rightarrow \underset{\text{(red)}}{HgI_2} \xrightarrow{2I^-} \underset{\text{colourless}}{HgI_4{}^{2-}}$$

$$Hg^{+2} + Co^{+2} + 4SCN^- \rightarrow CO\left[Hg(SCN)_4\right] \downarrow$$

Q.4 Answer:

Acidic $AgNO_3$

Solution:

$$2AgNO_3 + H_2\ S \rightarrow \underset{\text{Black ppt.}}{Ag_2\ S} + 2HNO_3$$

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Q.5 Answer:

SO_3^{2-}

Solution:

$$Na_2SO_3 + 2HCl\ \textbf{(dil.)} \rightarrow 2NaCl + H_2O + SO_2$$

$$K_2Cr_2O_2 + H_2SO_4 + 3SO_4 \rightarrow K_2SO_4 + \underset{\text{(Green)}}{Cr_2(SO_4)_3} + H_2O$$

Q.6 Answer:

Brown ring test

Solution:

Brown ring test is not a preliminary test.

Q.7 Answer:

All

Solution:

$$Pb^{+2} + SO_4^{-2} \rightarrow \underset{\text{(white)}}{PbSO_4}\downarrow$$

$$Pb^{+2} + CrO_4^{-2} \rightarrow \underset{\text{(yellow)}}{PbCrO_4}\downarrow$$

$$Pb^{+2} + I^{-1} \rightarrow \underset{\text{(yellow)}}{PbI_2}\downarrow$$

Q.8 Answer:

S^{2-}

Solution:

$$ZnS \xrightarrow{dlH_2SO_4} \underset{\text{(B)}}{ZnSO_4} + H_2S$$

$$\underset{\text{(A)}}{H_2S} \xrightarrow{K_2Cr_2O} S^{2-}\downarrow + Cr^{3+}$$

Q.9 Answer:

$Sb_2\ S_3$

Solution:

$Sb_2\ S_3$ **is an acidic salt. It reacts with base like** $NaOH$.

$$Sb_2S_3 + 6NaOH \rightarrow Na_3AsO_3 + Na_3\left[A_sS_3\right] + 3H_2O$$

Q.10 Answer:

PbI_2 and $K_2\left[PbI_4\right]$ respectively

Solution:

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$$PbCl_2 + KI \longrightarrow \underset{\text{yellow ppt}}{PbI_2} + KCl$$

$$PbI_2 + KI \longrightarrow K_2[PbI_4]$$

Q.11 Answer:

NH_4SCN

Solution:

The chemical compound to be distinguished should gives different colouration with reagent.

$Fe^{2+} + NH_4SCN \rightarrow$ **No change in colour**

$Fe^{3+} + NH_4SCN \rightarrow$ **Red colouration**

Therefore, NH_4SCN is used to distinguish between Fe^{2+} and Fe^{3+}.

Q.12 Answer:

$Ag(NH_3)_2^+$ and Cl^-

Solution:

$$AgCl + 2NH_3 \rightarrow \underset{\text{complex}}{[Ag(NH_3)_2]Cl} \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$$

Q.13 Answer:

Na_2CO_3

Solution:

$$\underset{(X)}{CaCO_3} \longrightarrow CaO + CO_2\uparrow$$

$$CaO + H_2O \longrightarrow \underset{Y}{Ca(OH)_2}$$

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2\uparrow$$

$$Cu^{+2} + 2CN^- \longrightarrow Cu(CN)_2$$

Q.14 Answer:

Zinc

Solution:

ZnS is white.

Q.15 Answer:

Sb_2S_5

Solution:

$$Sb_2S_3 + 2(NH_4)_2S_2 \rightarrow 2(NH_4)_2S + Sb_2S_5$$

Q.16 Answer:

Group I basic radicals precipitate as chlorides

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Solution:

Group I basic radical $\left(Pb^{2+}, Ag^{+}, {Hg_2}^{2+}\right)$ precipitate as chloride group IV basic radicals $\left(Ni^{+}, Co^{2+}, Mn^{2+}, Zn^{2+}\right)$ precipitate as sulphide group V basic radicals $\left(Ba^{2+}, Sr^{2+}, Co^{+}\right)$ precipitate as carbonate.

Q.17 Answer:

ZnS

Solution:

$$ZnS + H_2SO_4 \rightarrow H_2\,S + ZnsO_4$$
$$(\,A) \qquad\qquad (\,C)$$
$$H_2\,S + K_2Cr_2O_7 \rightarrow S + Cr_2O_3$$
$$S + O_2 \rightarrow SO_2$$
$$H_2\,S + SO_2 \rightarrow S + H_2O$$
$$CuSO_4 + H_2O \rightarrow \underset{(\text{Blue})}{CuSO_4 5H_2O}$$
$$ZnSO_4 + NaOH \rightarrow Zn(OH)_2 \xrightarrow{NaOH} Na_2ZnO_2$$

Q.18 Answer:

$H_2\,S$

Solution:

$$ZnS + H_2SO_4 \rightarrow H_2\,S + ZnsO_4$$
$$(\,A) \qquad\qquad (\,C)$$
$$H_2\,S + K_2Cr_2O_7 \rightarrow S + Cr_2O_3$$
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$$H_2\,S + SO_2 \rightarrow S + H_2O$$
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$$ZnSO_4 + NaOH \rightarrow Zn(OH)_2 \xrightarrow{NaOH} Na_2ZnO_2$$

Q.19 Answer:

$ZnSO_4$

Solution:

$$ZnS + H_2SO_4 \rightarrow H_2\,S + ZnsO_4$$
$$(\,A) \qquad\qquad (\,C)$$
$$H_2\,S + K_2Cr_2O_7 \rightarrow S + Cr_2O_3$$
$$S + O_2 \rightarrow SO_2$$
$$H_2\,S + SO_2 \rightarrow S + H_2O$$
$$CuSO_4 + H_2O \rightarrow \underset{(\text{Blue})}{CuSO_4 5H_2O}$$
$$ZnSO_4 + NaOH \rightarrow Zn(OH)_2 \xrightarrow{NaOH} Na_2ZnO_2$$

Q.20 Answer:

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$Pb(NO_3)_2$

Solution:

As bottle $2+$ bottle 3 gives colourless and odourless gas, it may be carbon dioxide.

Generally carbonates are decomposed by acids giving CO_2 gas. It suggests that bottle 2 and 3 contains sodium carbonate and HCl.

Bottle $3+4$ gives blue precipitate which confirms the

Cu^{2+} in either of bottles,

$CuSO_4, CuCl_2$ and $Cu(NO_3)_2$ are

soluble and $CuCO_3$ is insoluble in water as evident from the reaction.

$Cu^{2+} + CO_3^{2-} \rightarrow CuCO_3 \downarrow$ (blue).Thus blue precipitate must be of copper carbonate.

Hence, bottle 4 is $CuSO_4$, 3 is Na_2CO_3, 2 is HCl (from above) and 1 is $Pb(NO_3)_2$ as it gives white precipitate of $PbCl_2$ with bottle (2).

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